

PRESIDENCY UNIVERSITY

Department of Computer Science and Engineering

CSE321

MICROPROCESSORS BASED SYSTEM DESIGN

THEORY COURSE

Ten-Lecture Course Plan

Course Code	CSE321
Course Type	Theory
Credit Hours	03
Prerequisite	CSE207
Number of Lectures	10
Primary Focus	8086 Architecture, Assembly, Memory and Peripheral Interfacing

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1 Course Overview

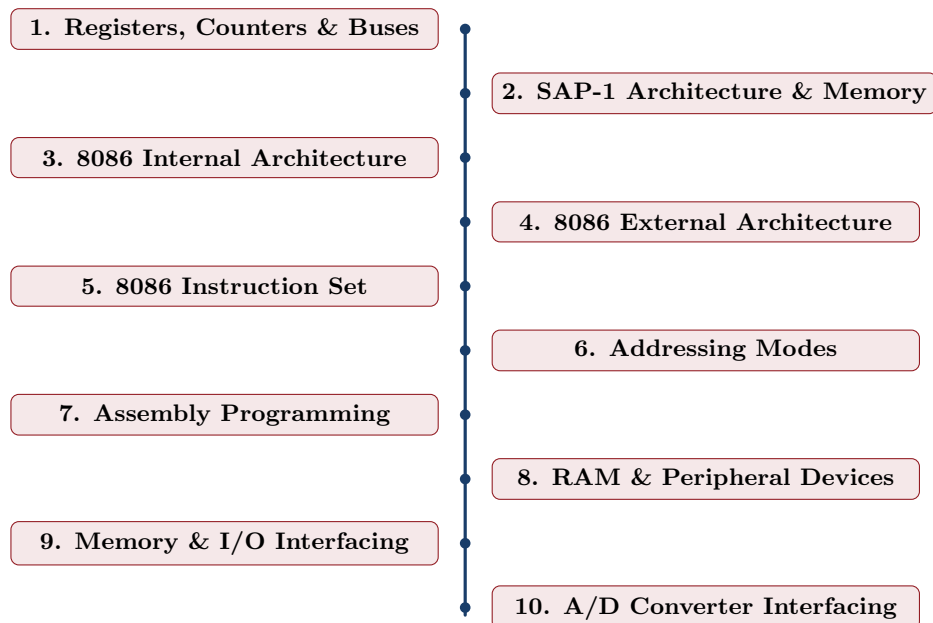
CSE321: Microprocessors Based System Design is a three-credit theory course following **CSE207**. A microcomputer uses a microprocessor as its central processing unit, with data-processing and control functionality integrated on a single chip. This course builds the theoretical foundation required to understand processor organization, write 8086 assembly programs, and design memory and peripheral interfaces.

Course rationale. Students progress from digital data-path components to the SAP-1 teaching processor, then to the architecture of the Intel 8086 and its CISC instruction set. They subsequently analyze software execution, RAM organization, digital peripheral connections, and analog-to-digital conversion interfaces. Theory classes emphasize worked calculations, instruction traces, timing diagrams, logic diagrams, and written interface designs; physical experimentation belongs to a separate laboratory course.

Course description coverage. Review of registers, counters, and buses; SAP-1 architecture and memory organization; internal and external architecture of the Intel 8086; its CISC instruction set and addressing modes; assembly-language semantics and implementation; basic RAM and peripheral architectures; memory and peripheral interfacing; and A/D converter interfacing.

1.1 Course Progression

The course preserves the order of the supplied description, with A/D conversion added at the end to meet CO4.



2 Course Objectives

The course aims to enable students to:

1. Relate registers, counters, buses, ALUs, control units, and memory to the operation of a stored-program computer.
2. Trace SAP-1 instructions and explain the connection between a datapath, control signals, and instruction execution.

3. Explain 8086 internal blocks, pin-level behavior, memory segmentation, bus cycles, and address generation.
4. Classify and use 8086 instructions and addressing modes, and construct traceable assembly-language solutions.
5. Analyze RAM organization, bank selection, address decoding, and digital peripheral interfacing with the 8086.
6. Design and explain an 8086-to-ADC interface, including conversion control, polling, digital-data transfer, and result interpretation.

3 Course Outcomes (COs)

At the end of the course, students will be able to:

CO	Supplied Course Outcome
CO1	Explain the Intel 8086 Microprocessor Architecture.
CO2	Understand and Write 8086 Assembly Language Programs.
CO3	Explain and Implement the Interfacing of Microprocessor with Peripheral Devices.
CO4	Explain and Implement the Interfacing with A/D converters.

Interpretation in a theory course. The word *implement* in CO3 and CO4 is addressed through correctly annotated interface circuits, port assignments, control/timing sequences, and compatible assembly routines. Hardware testing is not assumed in the theory sessions.

4 Delivery Structure and Equal Allocation

Equal division: The plan consists of **10 lectures**; each lecture receives **10% of the ten-lecture teaching unit** and contains **four blocks of 25% of that lecture**. Thus the plan allocates exactly **40 equal teaching blocks** without presuming a particular class duration. The cognitive load varies by topic, so selected instruction variants and lengthy exercises can be consolidated through reading and short take-home practice.

Lec.	Topic	Four-block emphasis	CO
1	Digital components and microprocessor fundamentals	Registers; counters; buses; CPU/data path and instruction cycle	Base
2	SAP-1 architecture and memory	Blocks; register transfers; memory/instructions; control and tracing	Base
3	8086 internal architecture	CISC/BIU/EU; registers; flags/queue; segmentation/address calculation	CO1
4	8086 external architecture	Pins; multiplexing; minimum/maximum modes; bus cycles/interrupt signals	CO1
5	8086 instruction set	Formats/data transfer; arithmetic; logic/shift; branch/stack/string/I/O	CO2
6	8086 addressing modes	Immediate/register/direct; indirect/index; effective addresses; branches/ports	CO2

Lec.	Topic	Four-block emphasis	CO
7	Assembly language and execution	Syntax/directives; program structure; core algorithms; stack/procedures/debugging	CO2
8	RAM and peripheral architectures	Memory types; RAM internals; 8086 banks; device/register organization	CO3
9	Memory and peripheral interfacing	Decode and mapping; RAM/ROM design; I/O mapping; 8255 applications	CO3
10	A/D converter interfacing	Conversion basics; ADC0804; ADC-8255-8086 circuit; control program	CO4

5 Detailed Ten-Lecture Theory Plan

5.1 Lecture 1: Review of Digital Components and Microprocessor Fundamentals

Primary alignment: Foundational for CO1–CO4

Four equal teaching blocks

Learning target: Identify the functional units of a simple CPU and explain the movement of data and instructions.

Block 1 (25% of lecture) — Registers and storage

- Review flip-flops; storage registers and shift registers; register widths.
- Register-transfer notation, parallel loading, and control of data movement.

Block 2 (25% of lecture) — Counters and sequencing

- Binary counters, incrementing/counting operations, synchronous control.
- The program counter as the source of the next instruction address.

Block 3 (25% of lecture) — Bus organization

- Data, address, and control buses; unidirectional versus bidirectional lines.
- Shared/common bus operation, selection and tri-state enabling; bus width.

Block 4 (25% of lecture) — From components to a microcomputer

- ALU, control unit, registers, memory, I/O, clock, and interconnecting buses.
- Basic instruction cycle: fetch, decode, execute, and storage of results.

Board-work / guided example: Draw a minimal CPU–memory–I/O block diagram and trace an instruction address from PC to memory.

Exit check: Distinguish the purpose and direction of the three system buses.

5.2 Lecture 2: SAP-1 Architecture and Memory Organization

Primary alignment: Foundation for CO1 and CO2

Four equal teaching blocks

Learning target: Trace elementary stored-program execution using SAP-1 blocks, memory, and control signals.

Block 1 (25% of lecture) — SAP-1 organization

- SAP-1 purpose and architectural block diagram; simplified 8-bit data path.
- Program counter, memory address register (MAR), instruction register, accumulator, B register, ALU, output register, and controller/sequencer.

Block 2 (25% of lecture) — Registers, buses, and transfers

- Shared W bus; register-enable/load controls and ALU add/subtract role.
- PC increment, instruction address transfer to MAR, and operand transfer.

Block 3 (25% of lecture) — Memory and instruction encoding

- SAP-1 16-word by 8-bit RAM: four-bit addresses and eight-bit data.
- Four-bit opcode plus four-bit address/operand field; LDA, ADD, SUB, OUT, HLT.

Block 4 (25% of lecture) — Control timing and execution

- Fetch timing states and control words/micro-operations; execution timing by opcode.
- Stepwise trace of a small load-add-output program including memory contents.

Board-work / guided example: Prepare a 16-location SAP-1 memory image for LDA, ADD, OUT, HLT and trace PC, MAR, IR, A, and output over timing states.

Exit check: Explain how the opcode selects control signals and why RAM addresses are four bits.

5.3 Lecture 3: 8086 Microprocessor: Internal Architecture

Primary alignment: CO1

Four equal teaching blocks

Learning target: Describe BIU/EU cooperation, register roles, flag semantics, and physical address formation.

Block 1 (25% of lecture) — 8086 and the CISC model

- 16-bit Intel 8086 overview, variable-length instruction set, and comparison with SAP-1.
- Execution Unit (EU) and Bus Interface Unit (BIU): functional responsibilities.

Block 2 (25% of lecture) — Register file and processor state

- General-purpose AX, BX, CX, DX and AH/AL byte halves; SP, BP, SI, DI.
- CS, DS, SS, ES; IP and the status/control FLAGS register.

Block 3 (25% of lecture) — Instruction pipeline and flags

- Six-byte prefetch instruction queue and overlapping bus access with execution.
- CF, PF, AF, ZF, SF, OF; TF, IF, DF; their use by arithmetic and control flow.

Block 4 (25% of lecture) — Segmented memory organization

- 20-bit physical address bus, 1 MiB space, segment:offset pair and 64 KiB segment range.
- Physical address = (segment $\times 10_{16}$) + offset; overlapping segments.

Board-work / guided example: Calculate the physical address for 1234h:5678h; label BIU and EU responsibilities during a memory operand instruction.

Exit check: Identify the default purposes of CS, DS, SS and ES and interpret ZF and CF.

5.4 Lecture 4: 8086 Microprocessor: External Architecture

Primary alignment: CO1

Four equal teaching blocks

Learning target: Explain the function of external pins, control signals, bus modes and bus timing.

Block 1 (25% of lecture) — External signal groups

- 8086 pin diagram; AD0–AD15, A16/S3–A19/S6, BHE#/S7 and other status pins.
- RD#, WR#, ALE, READY, RESET, CLK, NMI, INTR and INTA#; active-low notation.

Block 2 (25% of lecture) — Multiplexing and bus support

- Multiplexed address/data lines and why an address latch is needed.
- ALE-controlled demultiplexing, bus transceivers, DEN#, and DT/R#.

Block 3 (25% of lecture) — Operating modes

- MN/MX#-selected minimum and maximum modes and system configurations.
- Minimum-mode M/IO, RD#, WR# controls; maximum-mode S2–S0 signals and 8288 bus controller.

Block 4 (25% of lecture) — External transactions and arbitration

- Memory/I/O read and write bus cycles, T1–T4, READY-generated wait states.
- HOLD/HLDA (minimum mode), request/grant (maximum mode), interrupt acknowledgment.

Board-work / guided example: Sketch a simplified multiplexed-bus latch arrangement and label the T1–T4 memory-read cycle.

Exit check: Compare minimum versus maximum mode and explain READY and ALE.

5.5 Lecture 5: 8086 Instruction Set

Primary alignment: CO2

Four equal teaching blocks

Learning target: Classify 8086 instruction families and predict operand changes and flag effects.

Block 1 (25% of lecture) — Instruction representation and data movement

- Mnemonic/operands, byte versus word, instruction encoding idea and operand restrictions.
- MOV, XCHG, LEA, PUSH, POP, PUSHF/POPF; transfers among registers, memory, stack.

Block 2 (25% of lecture) — Arithmetic and comparisons

- ADD, ADC, SUB, SBB, INC, DEC, CMP, NEG; unsigned/signed interpretations.
- MUL/IMUL and DIV/IDIV operand/result conventions; decimal-adjust family overview.

Block 3 (25% of lecture) — Logic, shifts, and rotates

- AND, OR, XOR, NOT, TEST; bit masking and testing.
- SHL/SAL, SHR, SAR; ROL, ROR, RCL, RCR; relevant carry/sign/zero effects.

Block 4 (25% of lecture) — Flow, strings, I/O, and processor control

- JMP, signed/unsigned Jcc, LOOP, JCXZ, CALL, RET; stack linkage.
- MOVS, CMPS, SCAS, LODS, STOS with REP variants; IN, OUT, INT, IRET; CLC/STC, CLI/STI, CLD/STD, NOP, HLT.

Board-work / guided example: Trace `MOV AL,7Fh`; `ADD AL,01h`, identify resulting flags, then choose signed versus unsigned branches.

Exit check: Explain how CMP differs from SUB and name the instruction family for reading an I/O port.

5.6 Lecture 6: 8086 Addressing Modes

Primary alignment: CO2

Four equal teaching blocks

Learning target: Determine legal operand forms, effective addresses, default segments, and control-flow targets.

Block 1 (25% of lecture) — Basic operand addressing

- Immediate and register operands; direct memory displacement addressing.
- Operand-size identification (byte/word), register-only versus memory operands.

Block 2 (25% of lecture) — Register-indirect and combined addressing

- Indirect through BX, BP, SI, DI and legal base/index register combinations.
- Based, indexed, based-indexed, and displacement-based address forms.

Block 3 (25% of lecture) — Effective and physical address calculation

- Offset/effective address computations; BP normally implies SS, others normally DS.
- Segment override, physical address expression, LEA versus data read, illegal combinations.

Block 4 (25% of lecture) — Special address semantics

- IP-relative conditional jumps; near/far transfer distinction; implied operands and stack access.
- String source DS:SI and destination ES:DI, with override restrictions; immediate-port versus DX-port addressing for IN/OUT.

Board-work / guided example: Compute the address used by `MOV AX,[BP+SI+08h]` for given SS/BP/SI; compare with `[BX+SI+08h]`.

Exit check: Identify legal 8086 memory-address registers and explain a segment override.

5.7 Lecture 7: 8086 Assembly Language Semantics and Implementation

Primary alignment: CO2

Four equal teaching blocks

Learning target: Write structured 8086 assembly solutions, trace execution, and explain program translation.

Block 1 (25% of lecture) — Syntax, symbols, and assembler directives

- Labels, mnemonics, operand order, constants, comments, and symbolic names.
- DB, DW, EQU, SEGMENT/ENDS, ASSUME, END; instruction versus directive.

Block 2 (25% of lecture) — Complete program structure

- Code, data and stack segments; initialization of DS and use of the program entry point.
- Assemble, link, load and execute; source versus machine code; trace register/memory state.

Block 3 (25% of lecture) — Program-building patterns

- Addition/subtraction and comparisons; signed versus unsigned decisions.
- IF/ELSE, counter-controlled loops, arrays via SI/DI, basic string copying/scanning.

Block 4 (25% of lecture) — Stack, procedures, and program services

- SS:SP stack behavior, PUSH/POP, CALL/RET, simple procedures and register preservation.
- INT/IRET concept; software service calls as environment-specific examples; debugging workflow.

Board-work / guided example: Write a small MASM/TASM-style program to sum a byte array into a word result, with DS initialized and an explicit exit convention.

Exit check: Trace one CALL/RET stack change and find an incorrect branch or register-size mismatch.

5.8 Lecture 8: Brief Architecture of RAM and Peripheral Devices

Primary alignment: CO3

Four equal teaching blocks

Learning target: Explain memory-device organization and the 8086 byte-banked connection to peripheral controllers.

Block 1 (25% of lecture) — Memory-device principles

- Volatile/nonvolatile memory; RAM versus ROM; SRAM versus DRAM and refresh.
- Addressable locations, word width, capacity, access time and read/write controls.

Block 2 (25% of lecture) — Internal RAM architecture

- Cell array, row/column decoding, sense amplifiers, I/O drivers and chip-select function.
- Read versus write cycle behavior; selecting address, enabling output and storing data.

Block 3 (25% of lecture) — Memory from the 8086 viewpoint

- Byte addressing, 16-bit bus and the low/even and high/odd memory banks.
- A0, BHE# and their byte/word selection roles; aligned and unaligned word transfers.

Block 4 (25% of lecture) — Peripheral architecture basics

- Input/output peripheral functions, data/status/control registers and handshaking.
- 8255 PPI overview: ports A, B, C, control word and Mode 0/1/2 concepts.

Board-work / guided example: Use A0 and BHE# to select the correct bank(s) for a byte at an odd address and a word at an odd address.

Exit check: Compare SRAM with DRAM and identify the functions of PPI data/control ports.

5.9 Lecture 9: Interfacing 8086 with Memory and Peripheral Devices

Primary alignment: CO3

Four equal teaching blocks

Learning target: Construct address-decoding logic and 8086-compatible memory and I/O interface designs.

Block 1 (25% of lecture) — Memory-map design and address decoding

- Address-range allocation, chip-select generation and full versus partial decoding.
- Decoder/latch/buffer functions; mapping capacity and device address-pin count.

Block 2 (25% of lecture) — RAM/ROM interfacing with 8086

- Low/high bank connections, A0 and BHE# selection, read/write strobes and bus direction.
- Worked SRAM/ROM connection: map addresses, generate chip select, and trace byte/word access.

Block 3 (25% of lecture) — I/O interfacing choices

- I/O-mapped (isolated) versus memory-mapped I/O; separate address spaces and control signals.
- IN/OUT port forms, port address decoding, peripheral status polling, interrupt overview.

Block 4 (25% of lecture) — 8255 PPI-based application

- 8255 block/port diagram and Mode 0 control-word programming; Mode 1/2 handshaking overview.
- Port mapping on the 8086, switches/LEDs as example peripherals, and assembly I/O sequence.

Board-work / guided example: Design a two-bank SRAM map and assign 8255 low-byte I/O ports at 80h, 82h, 84h, 86h; explain the 8086-to-8255 address-bit connections.

Exit check: Distinguish memory-mapped from isolated I/O and show how a chip-select is decoded.

5.10 Lecture 10: A/D Converter Interfacing with 8086

Primary alignment: CO4

Four equal teaching blocks

Learning target: Design a working ADC control path and assembly-level conversion/read sequence.

Block 1 (25% of lecture) — Analog-to-digital conversion theory

- Sampling, quantization, resolution, reference/full-scale voltage and error.
- Conversion time and effective step size: $\Delta V = V_{FS}/2^n$; digital-code interpretation.

Block 2 (25% of lecture) — ADC internal and external operation

- Generic ADC block diagram and successive-approximation operation in outline.
- ADC0804 example: eight-bit data lines, CS#, RD#, WR#, INTR#, clock and reference arrangement.

Block 3 (25% of lecture) — Hardware interface via 8255

- ADC D0–D7 to 8255 Port A (input); Port C control/status lines; common ground, power and reference.
- Illustrative Mode 0 directions: Port A input, PC upper input for INTR#, PC lower output for WR#/RD#; 8255 control word 98h.

Block 4 (25% of lecture) — Conversion algorithm and program trace

- Initialize PPI; pulse WR# to start; poll INTR# for completion; assert RD#, read byte, restore controls.
- Assembly IN/OUT and TEST/JNZ sequence; translate sample count to approximate analog voltage.

Board-work / guided example: Draw the 8086–8255–ADC0804 connection, label control-signal polarity, and pseudocode/trace one sample conversion.

Exit check: For an 8-bit 0–5 V converter, calculate the nominal step size and the meaning of a digital output of 128.

6 Course Outcome Alignment and Evidence

CO	Main lectures	Suitable evidence in a theory assessment
CO1	3–4	Draw and explain BIU/EU, registers, segmented address calculations, minimum/maximum modes, and a read/write bus cycle.
CO2	5–7	Identify correct instructions and addressing modes, write assembly programs, and trace operands/flags/stack changes.
CO3	8–9	Analyze RAM organization; generate an address map, bank selects and chip enables; devise a 8255 interface and I/O program.
CO4	10	Draw an ADC–PPI–8086 circuit; explain conversion control, polling, sample transfer and resolution calculations.

7 Explicit Course Description Coverage Audit

Syllabus element	Lecture	Status
Registers, counters, buses, and related CPU components	1	Covered
SAP-1 architecture and memory organization	2	Covered
8086 internal architecture and CISC ISA concepts	3	Covered
8086 external architecture and bus organization	4	Covered
8086 instruction set	5	Covered
8086 addressing modes	6	Covered
8086 assembly-language semantics and implementation	7	Covered
Brief architecture of RAM and peripheral devices	8	Covered
Interfacing with memory systems	9	Covered
Interfacing with peripheral devices	9	Covered
Interfacing with A/D converters (CO4)	10	Covered

8 Suggested Theory Teaching and Checking Strategy

Within every lecture, use a short prerequisite recall, explain block-level concepts, derive at least one board example, and close with an independently answerable check. The four blocks are equal in allocation; the checks and demonstrations are integrated into those blocks rather than treated as extra lectures.

- **Architecture (Lectures 1–4):** label block and pin diagrams, perform control-signal traces and calculate addresses.
- **Instruction and programming work (Lectures 5–7):** predict register/FLAGS changes and develop short annotated algorithms.
- **Interfaces (Lectures 8–10):** build address maps, read timing diagrams, design decoder logic and describe port-control sequences.

Workload note. Lectures 5 and 7 are dense. Keep the taught core focused on representative instructions, one well-explained program and one guided trace; provide additional forms and variants as reading or practice, avoiding superficial coverage of every opcode.

9 Suggested Practice and Formative Questions

1. **L1:** Build a control/data/address-bus diagram for a CPU that reads an instruction from RAM.
2. **L2:** Encode a SAP-1 addition program, populate RAM and trace all register changes.
3. **L3:** Compute two segment:offset physical addresses and explain queue prefetch behavior.
4. **L4:** Contrast minimum/maximum modes and label a memory-write timing diagram.
5. **L5:** Select instructions for signed comparison, bit masking and stack transfer; justify flag use.
6. **L6:** Classify addressing modes and find effective and physical addresses for valid operands.
7. **L7:** Write and trace an 8086 loop to find a maximum in an array; state initial assumptions.
8. **L8:** Explain why a word read at an odd byte address may require two 8086 bus cycles.

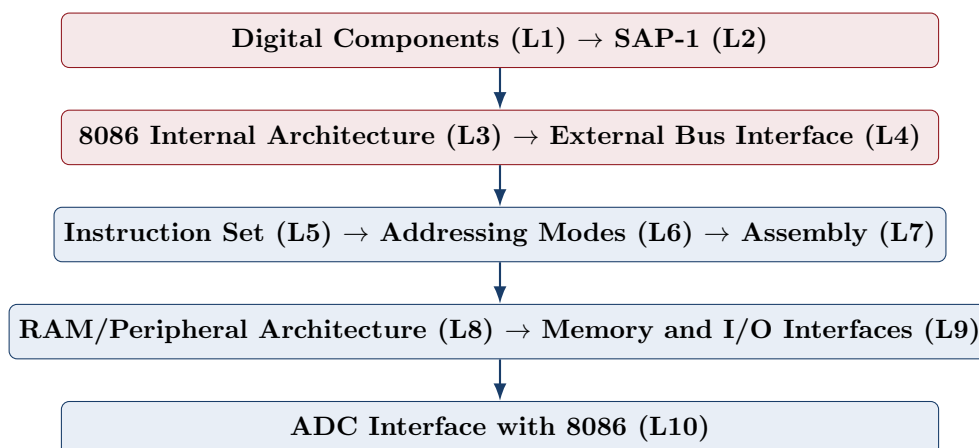
9. **L9:** Produce an address map and chip-enable expressions for RAM and an 8255-controlled output.
10. **L10:** Propose a polling-based ADC program and calculate quantization step size and sample voltage.

10 Recommended Books

1. Mohamed Rafiquzzaman (1995), *Microprocessors and Microcomputer-Based System Design*, 2nd edition, CRC Press.
2. Douglas V. Hall (1986), *Microprocessors and Interfacing: Programming and Hardware*, 1st edition, McGraw-Hill.

Reference use: the architectural and 8086 chapters support Lectures 1–7; the memory, I/O and peripheral-interface chapters support Lectures 8–10. Exact chapter/page assignments should be set against the edition available to students.

11 Course Concept Map



Conclusion

CSE321 has been divided into **ten equal lectures** with four consistent teaching blocks per lecture. The sequence begins with registers, counters and buses; builds understanding through SAP-1 and the 8086; then advances through instructions, addressing and programs to RAM, peripherals, memory interfaces and A/D converters. All elements of the supplied description and all four outcomes are explicitly mapped.